DERIVED LIMITS OF SURJECTIVE INVERSE SYSTEMS OF QUASI-COHERENT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof verifies on affine objects the vanishing of higher cohomology of quasi-coherent sheaves and the surjectivity of sections under surjective maps, then applies the two Cohomology on Sites lemmas for the three assertions

Editorial note. The label lemma-Rlim-quasi-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- vanishing of higher cohomology of quasi-coherent sheaves on affines (Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero) via the higher direct image discussion of Cohomology of Spaces
- surjectivity of H^0 on affines for surjective maps of quasi-coherent sheaves
- the derived limit criterion for systems of sheaves with surjective transitions and vanishing higher cohomology on a basis (Cohomology on Sites, Lemma sites-cohomology-lemma-derived-limit-suitable-system)
- the identification of ordinary and derived limits under these conditions (Cohomology on Sites, Lemma sites-cohomology-lemma-inverse-limit-is-derived-limit)

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Prerequisite pointer invoked at proof line(s) 608; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A09: sources/stacks-chow/tags.txt, source lines 12627–12627. Prerequisite pointer invoked at proof line(s) 612; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BKY: sources/stacks-chow/tags.txt, source lines 14576–14576. Prerequisite pointer invoked at proof line(s) 616; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 071Y: sources/stacks-chow/tags.txt, source lines 9007–9007. Prerequisite pointer invoked at proof line(s) 605; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let (K_n) be an inverse system of $D_{QCoh}(\mathcal{O}_X)$ with derived limit $K = R\lim K_n$ in $D(\mathcal{O}_X)$. Assume $H^q(K_{n+1}) \to H^q(K_n)$ is surjective for all $q \in \mathbf{Z}$ and $n \geq 1$. Then*

(1) *$H^q(K) = \lim H^q(K_n)$,*
(2) *$R\lim H^q(K_n) = \lim H^q(K_n)$, and*
(3) *for every affine open $U \subset X$ we have $H^p(U, \lim H^q(K_n)) = 0$ for $p > 0$.*

Proof. Let $\mathcal{B} \subset \mathrm{Ob}(X_{étale})$ be the set of affine objects. Since $H^q(K_n)$ is quasi-coherent we have $H^p(U, H^q(K_n)) = 0$ for $U \in \mathcal{B}$ by the discussion in Cohomology of Spaces, Section 071Y and Cohomology of Schemes, Lemma 01XB. Moreover, the maps $H^0(U, H^q(K_{n+1})) \to H^0(U, H^q(K_n))$ are surjective for $U \in \mathcal{B}$ by similar reasoning. Part (1) follows from Cohomology on Sites, Lemma 0A09 whose conditions we have just verified. Parts (2) and (3) follow from Cohomology on Sites, Lemma 0BKY. □